

E-Commerce Customer Churn Prediction

Machine Learning Final Project

Romaisaa Mohamed | Student No. 22100583

5,630

Rows

20

Columns

16.8%

Churn Rate

Problem Statement

Why This Matters

Acquiring a new customer costs **5–7x more** than retaining an existing one.

By predicting churn early, the business can proactively offer discounts or improve service for at-risk customers before they leave.

Research Questions

- 1 Can we predict whether a customer will churn?
- 2 Which behaviors and demographics are the strongest indicators?
- 3 Can we identify at-risk customers early enough to act?

Dataset & Data Cleaning

Dataset Info

Source	Kaggle — E-Commerce Churn
Rows	5,630
Columns	20
Target	Churn (0 = No, 1 = Yes)
Format	Excel (.xlsx), 2 sheets

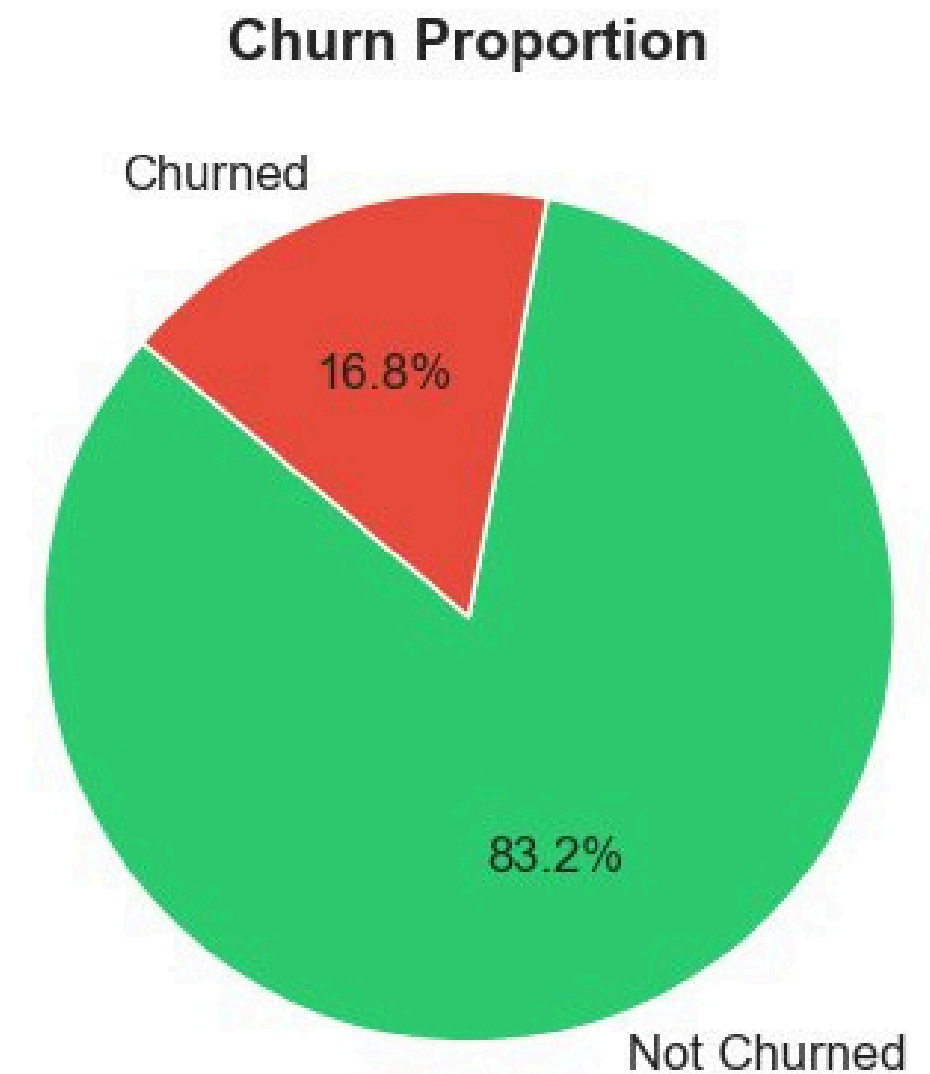
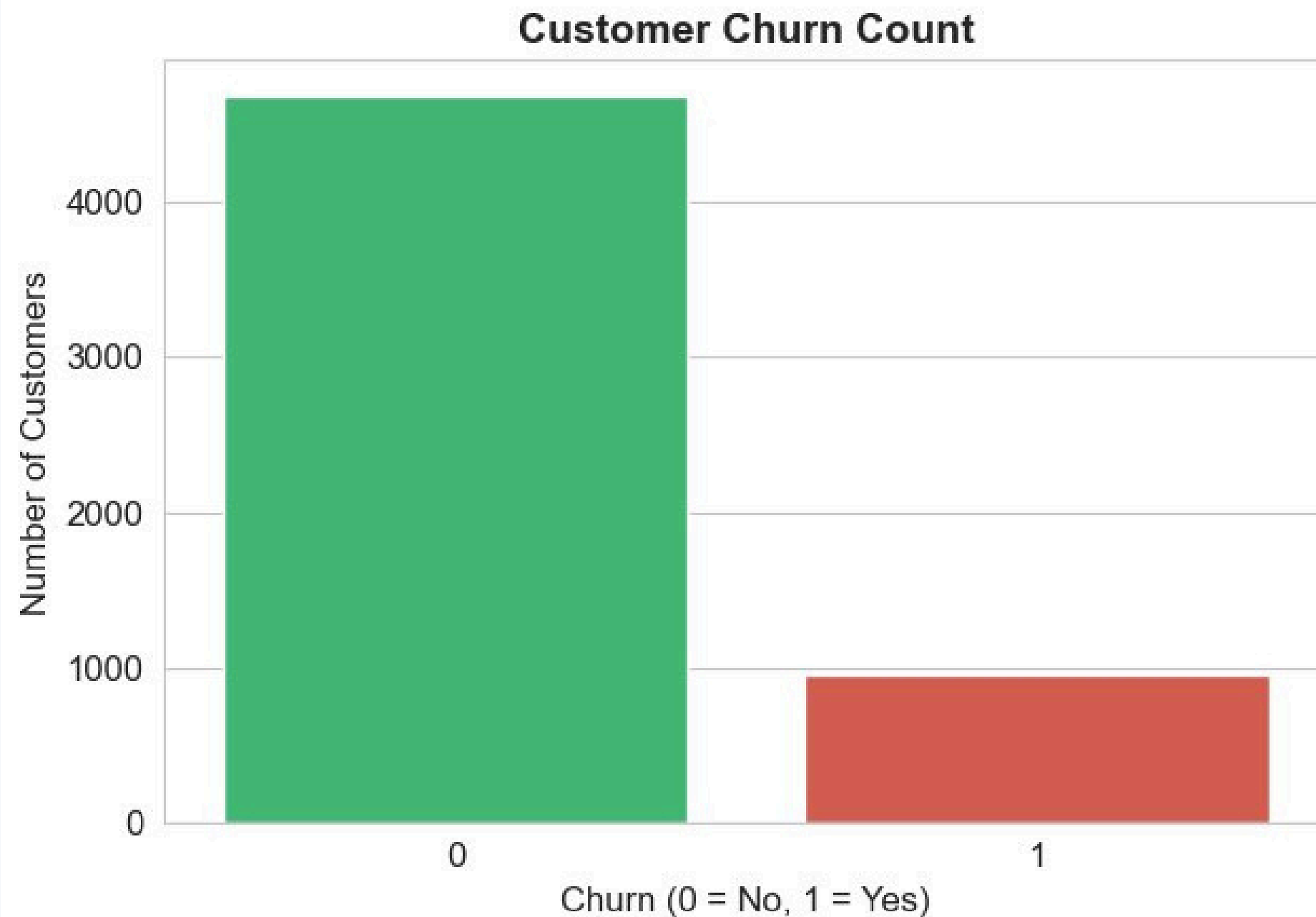
Cleaning Steps

- ✓ Loaded 'E Comm' sheet from Excel file
- ✓ Filled missing numeric values with median
- ✓ Filled missing categorical values with mode
- ✓ Removed duplicate rows
- ✓ Dropped CustomerID (not predictive)
- ✓ Encoded categorical columns with LabelEncoder

Exploratory Data Analysis — Churn Distribution

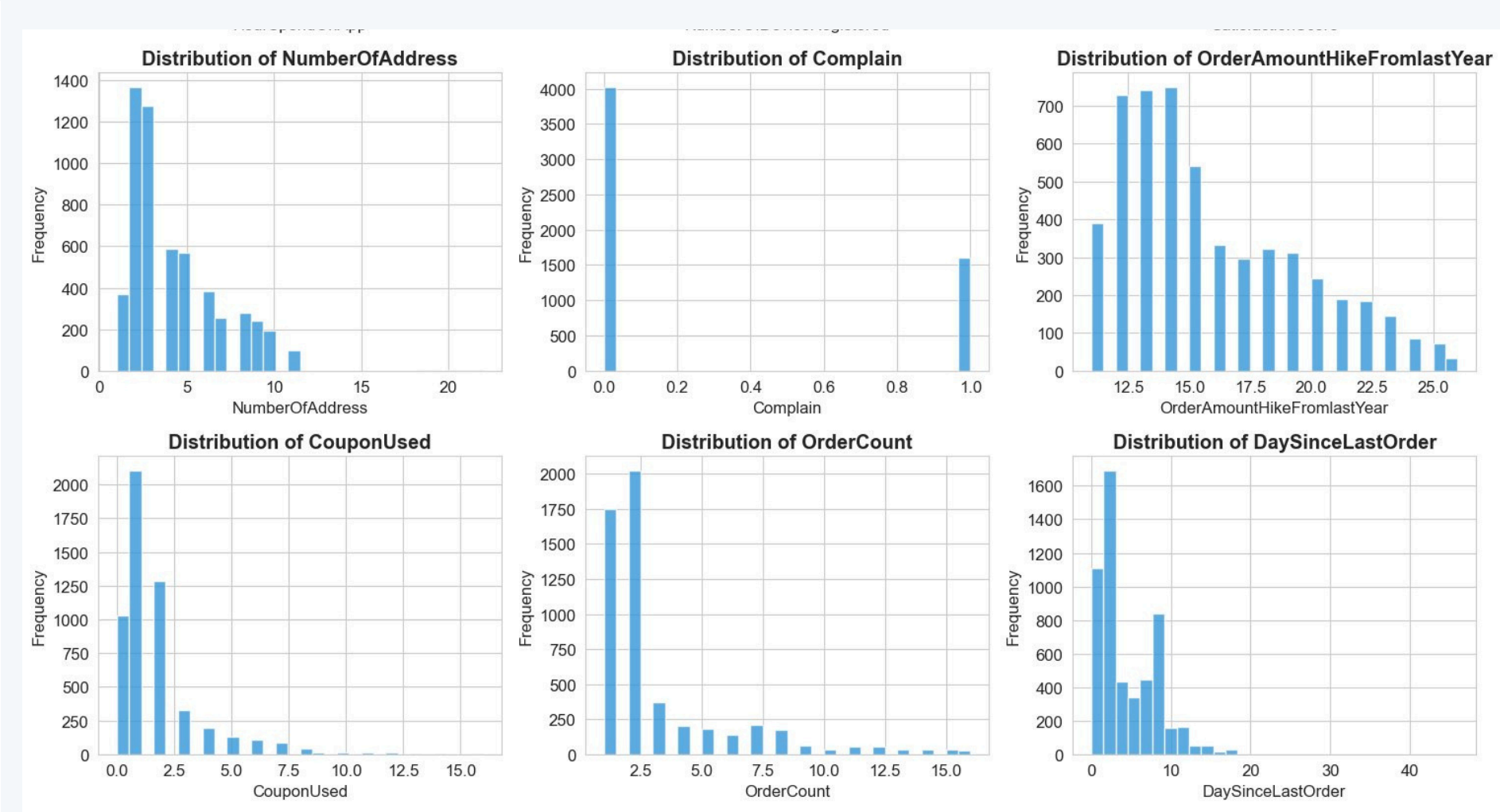
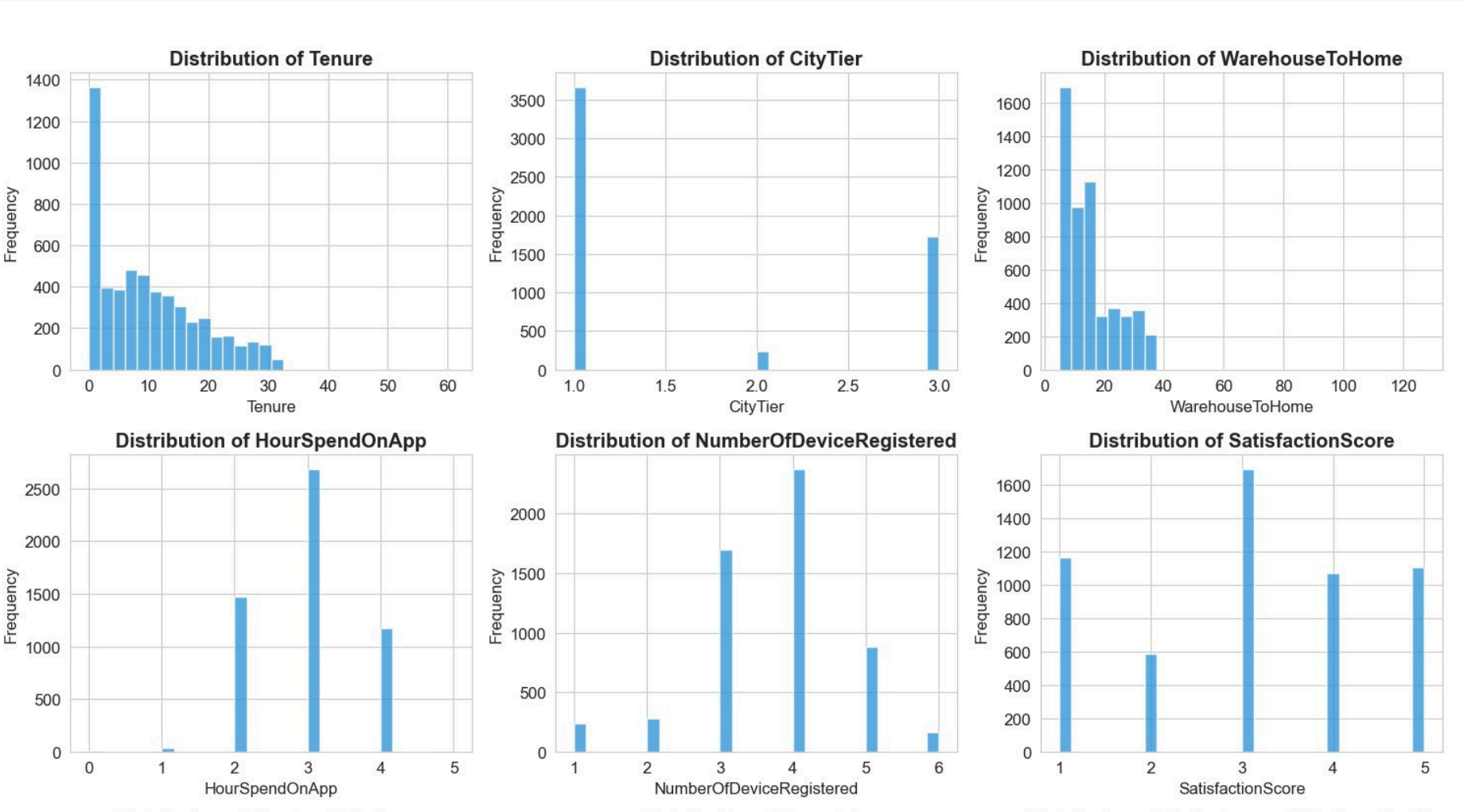
83.2% of customers stayed while 16.8% churned — the dataset is imbalanced.

Plot 1 & 2: Target Variable Distribution



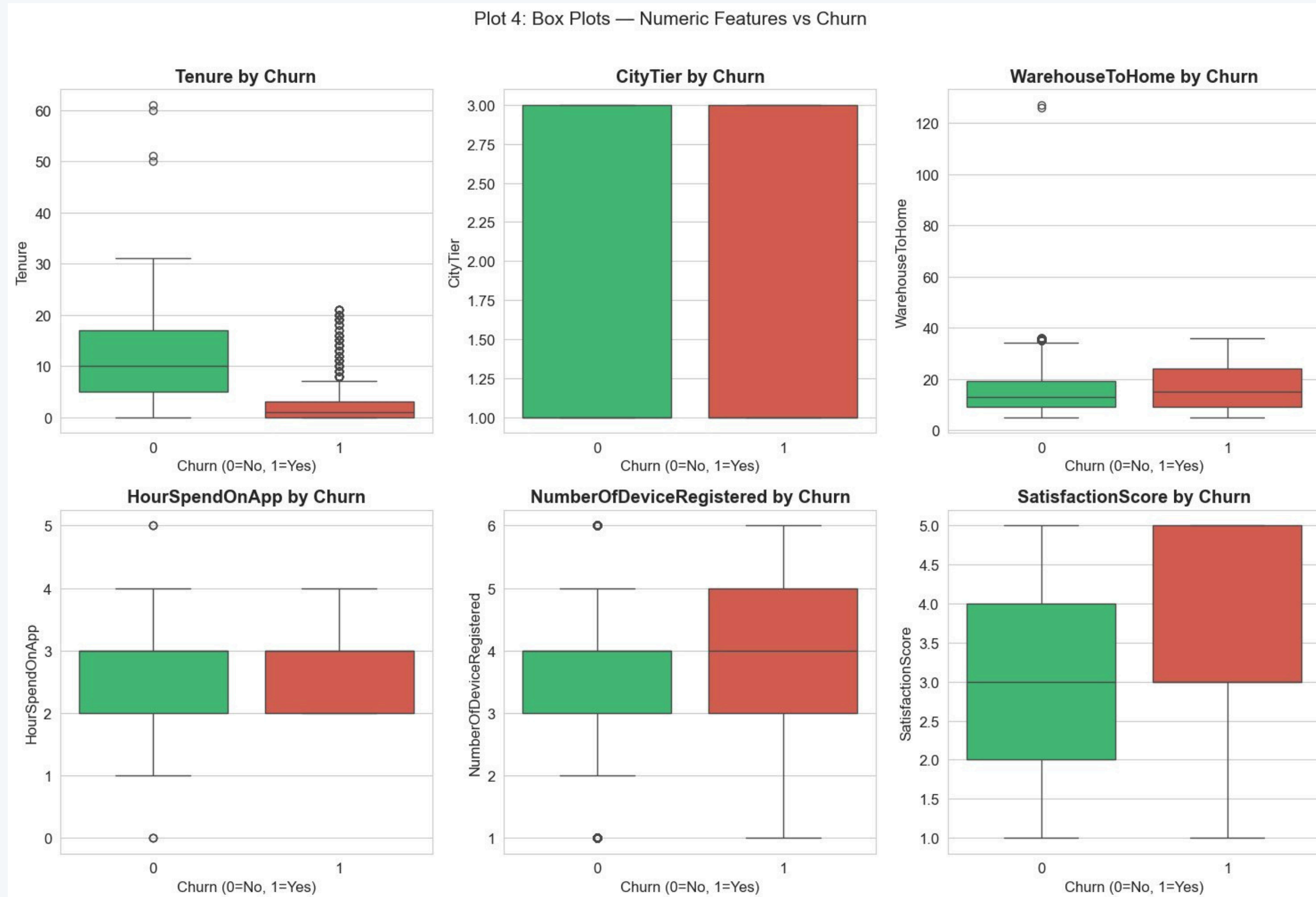
Exploratory Data Analysis — Numeric Distributions

Histograms reveal that Tenure is right-skewed (most customers are newer) and CashbackAmount is approximately normal.



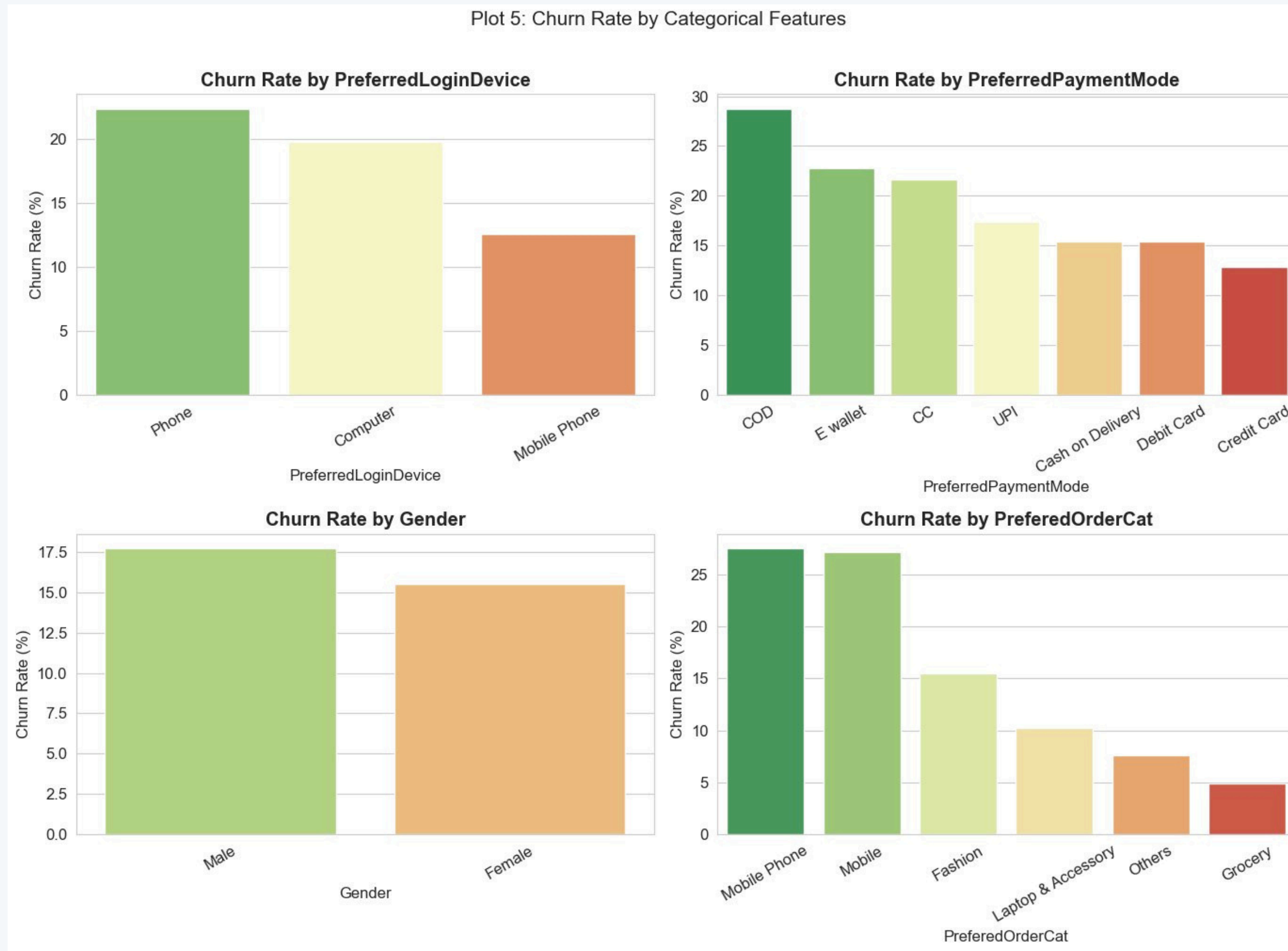
Exploratory Data Analysis — Box Plots vs Churn

Churned customers (1) have significantly lower Tenure and higher NumberOfDeviceRegistered than non-churned customers.



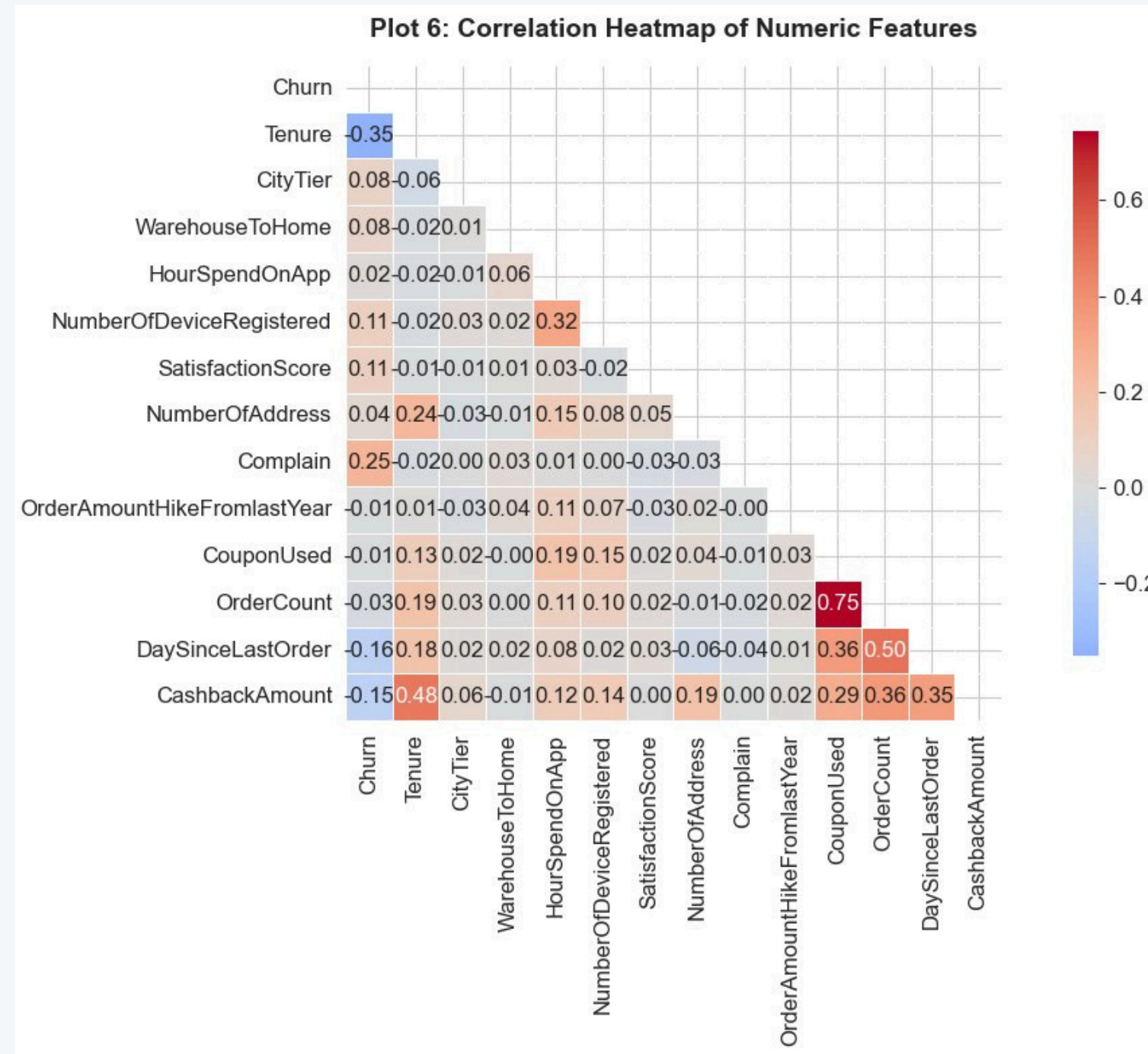
Exploratory Data Analysis — Churn by Category

Customers using Cash on Delivery and those ordering Mobile Phones have the highest churn rates.

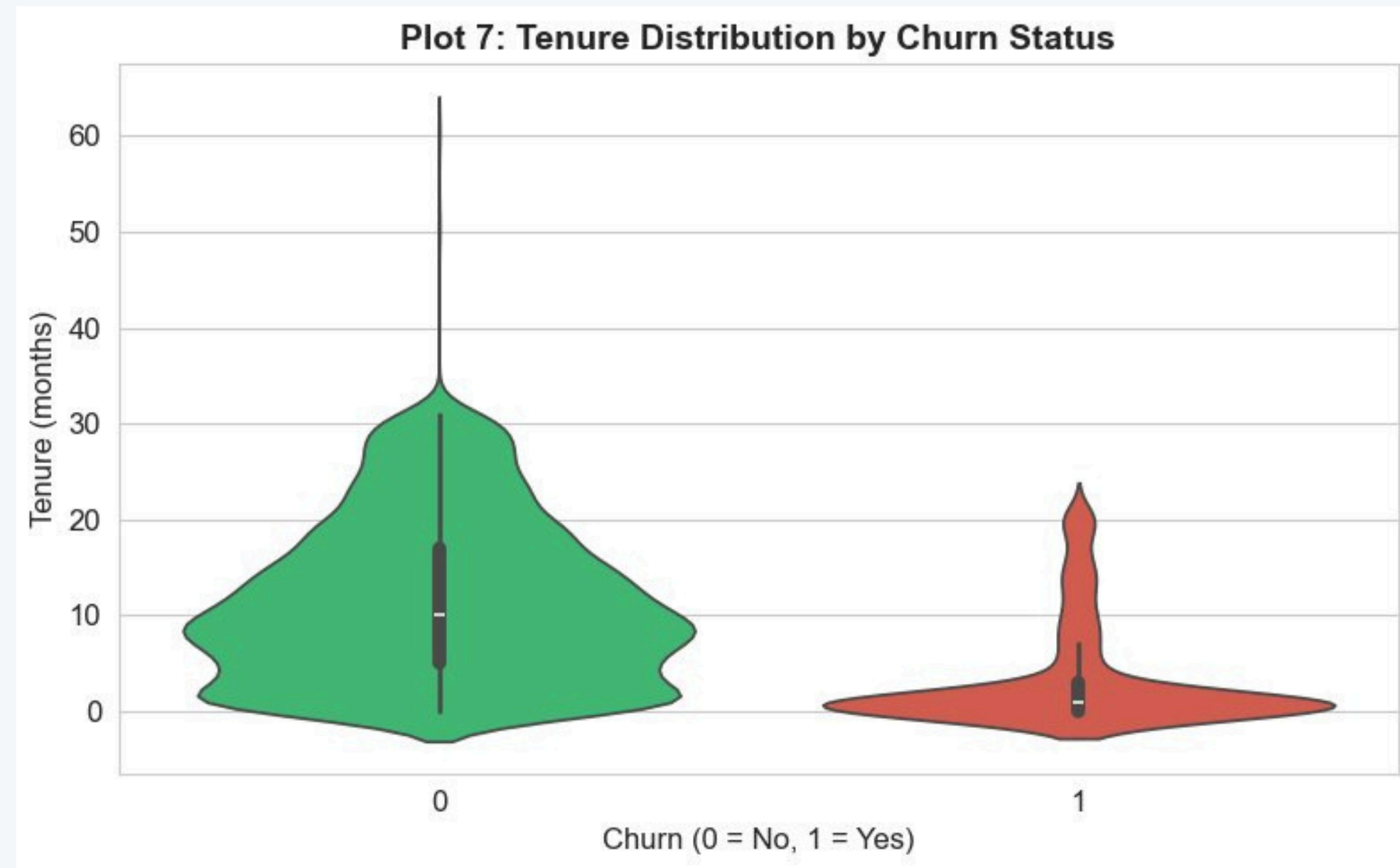


Exploratory Data Analysis — Correlation & Tenure

Heatmap



Violin — Tenure by Churn



Tenure is the strongest negative correlate of churn (-0.35). New customers churn at much higher rates.

Feature Engineering

EngagementScore

$$\text{Tenure} \div (\text{DaySinceLastOrder} + 1)$$

Measures how recently active a customer is. Higher = more engaged.

ComplaintsPerOrder

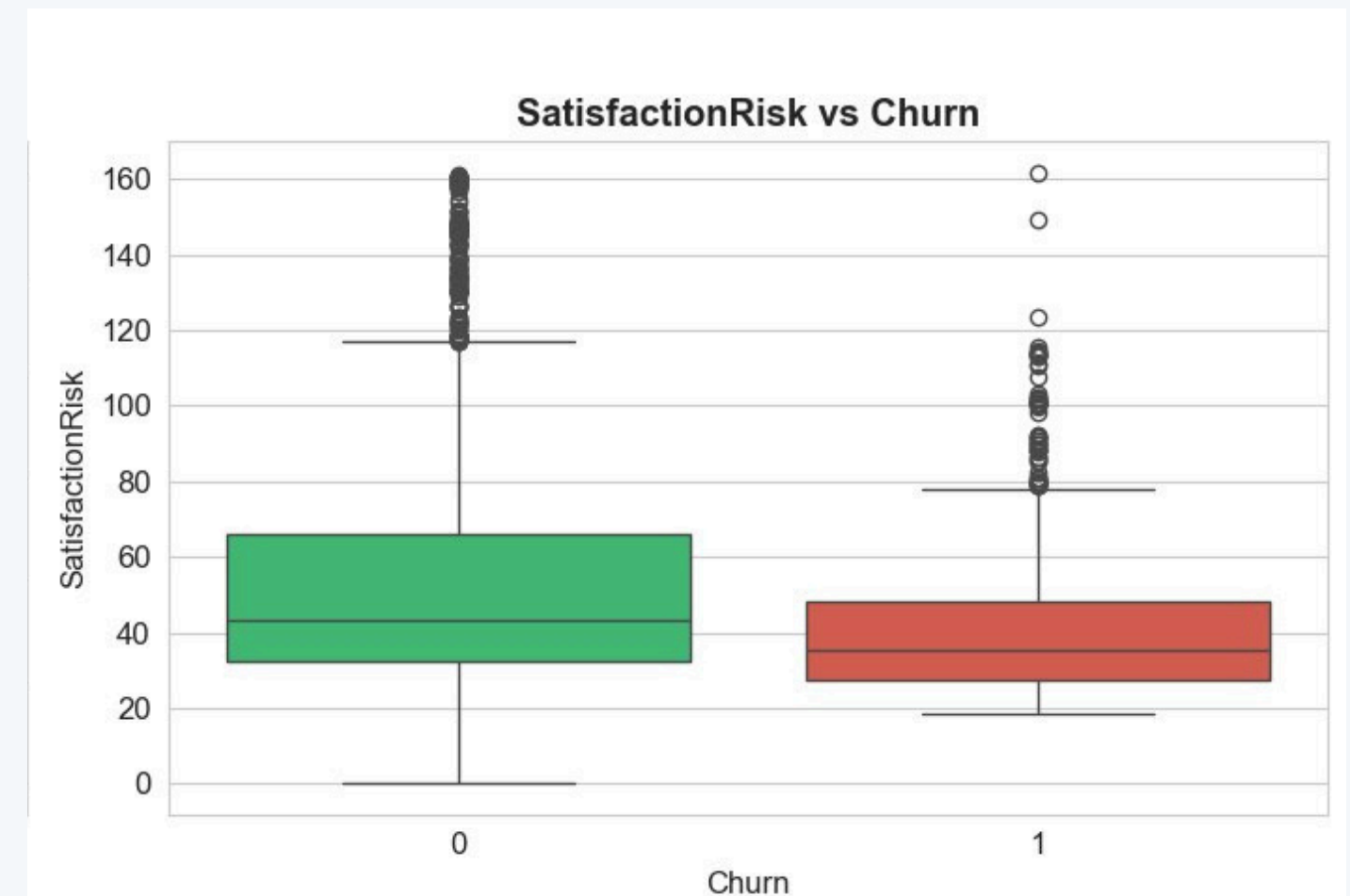
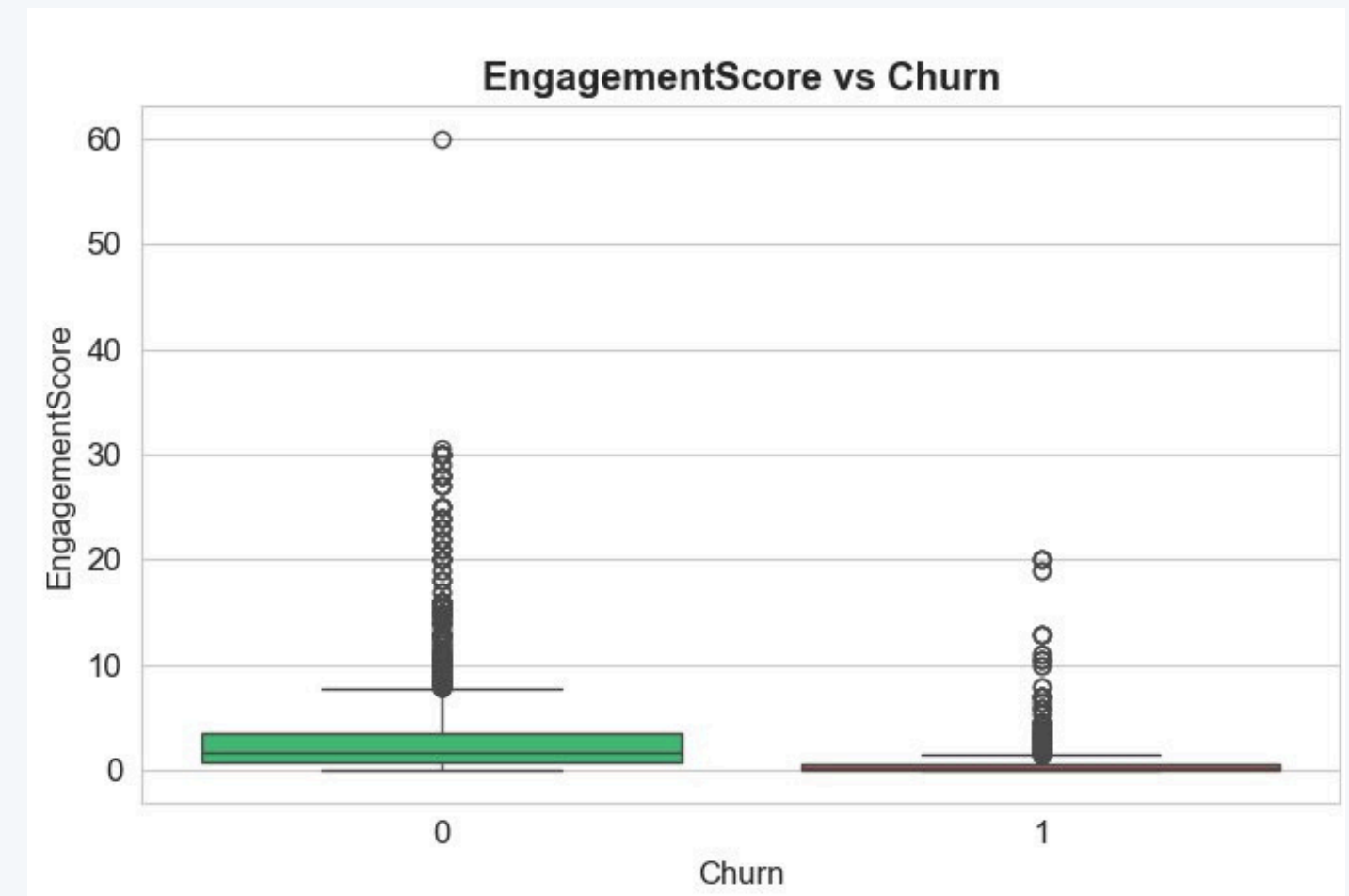
$$\text{Complain} \div (\text{NumberOfOrders} + 1)$$

Normalises complaint frequency relative to order volume.

SatisfactionRisk

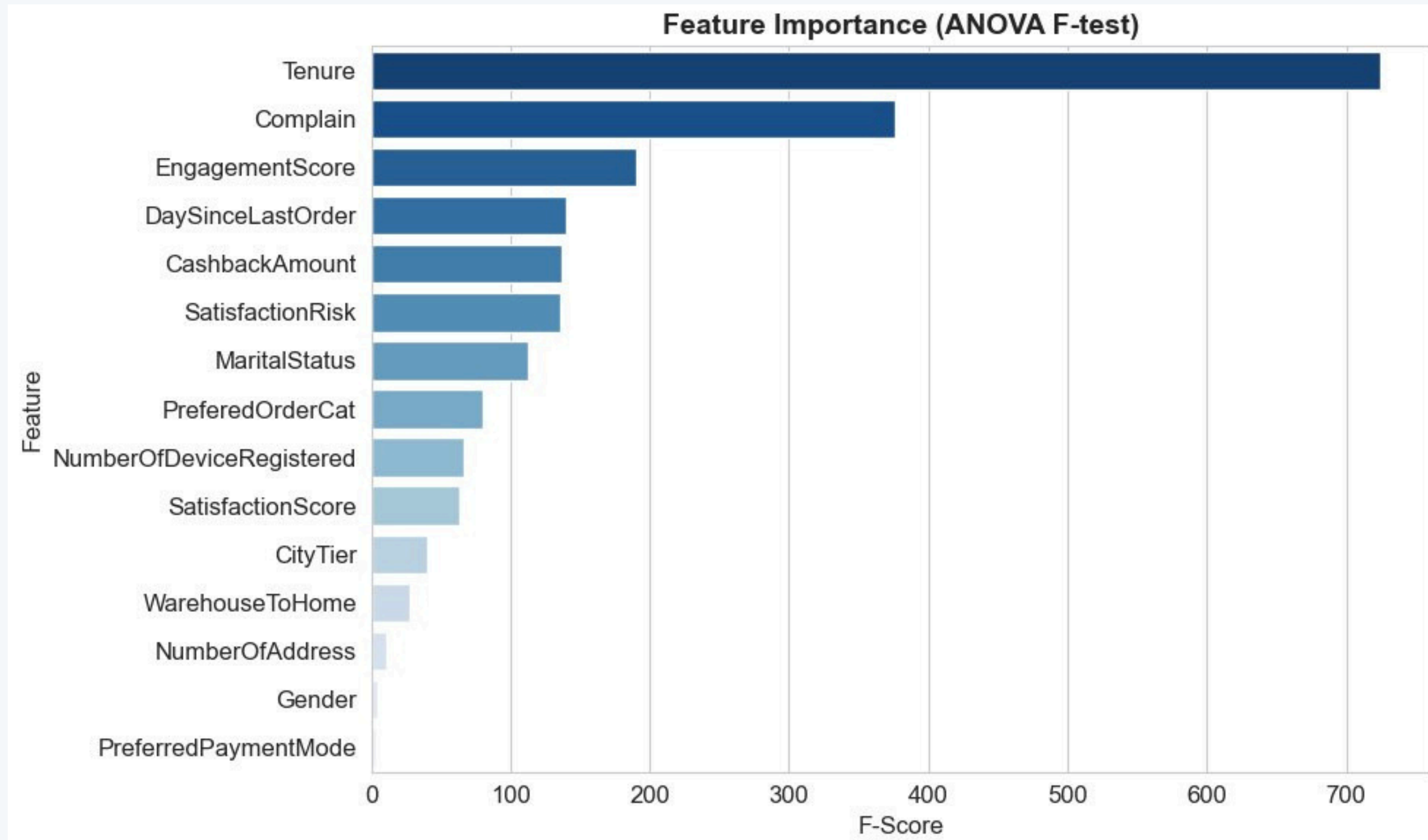
$$\text{CashbackAmount} \div (\text{SatisfactionScore} + 1)$$

Flags customers receiving high cashback despite low satisfaction.

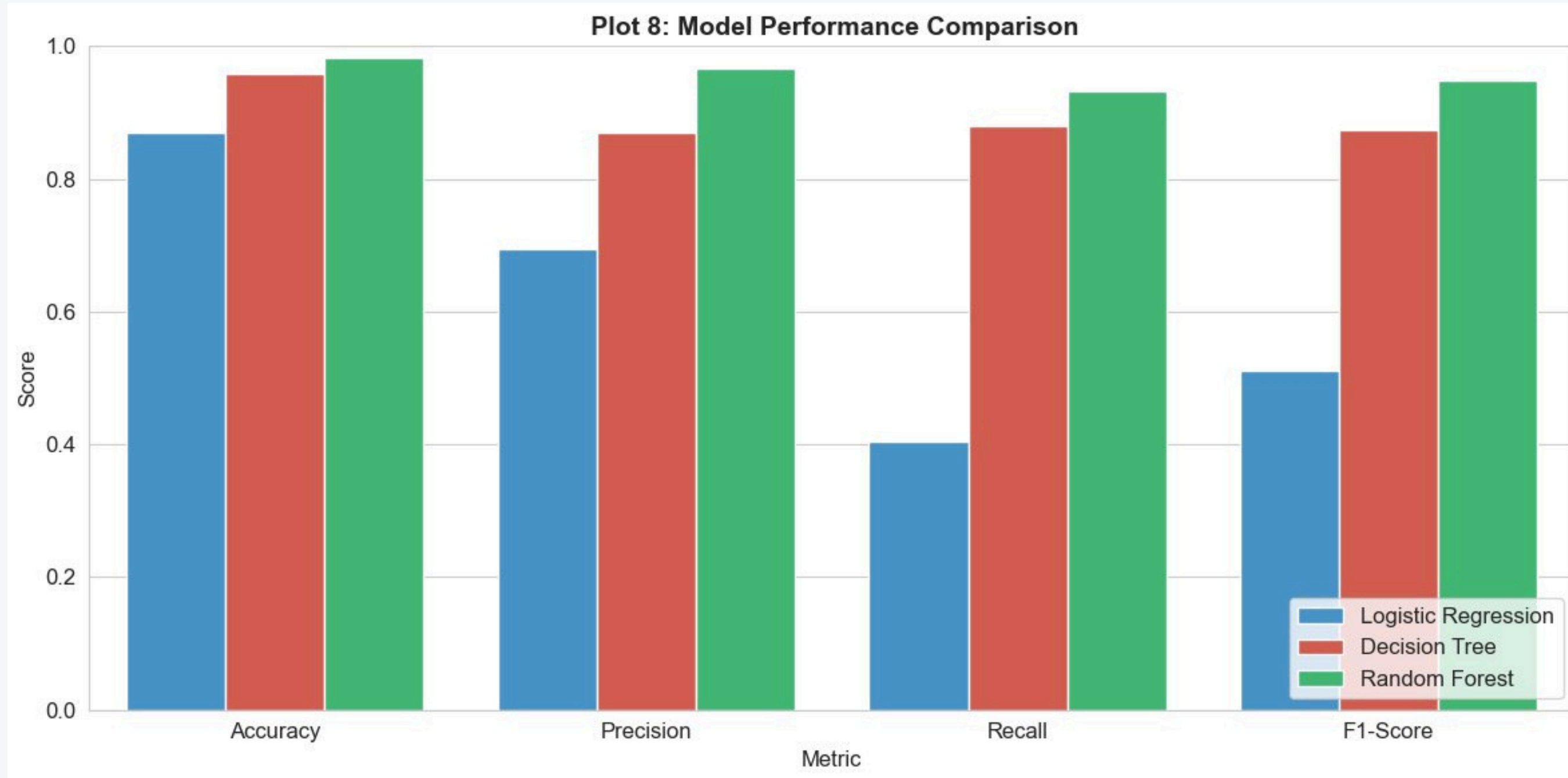


Feature Selection — Filter Method (ANOVA F-test)

SelectKBest scores each feature by how well it separates the two classes. Top 15 features were kept.



Model Comparison — 3 Algorithms



Random Forest selected as the final model based on highest performance across all metrics.

Parameter Tuning — GridSearchCV

What is Parameter Tuning?

Default algorithm settings are just a starting guess.
Tuning finds the optimal combination of parameters to maximise model performance using cross-validation.

Parameters Tuned

- n_estimators: [100, 200]
- max_depth: [None, 10, 20]
- min_samples_split: [2, 5]
- class_weight: [balanced, None]

GridSearchCV Results

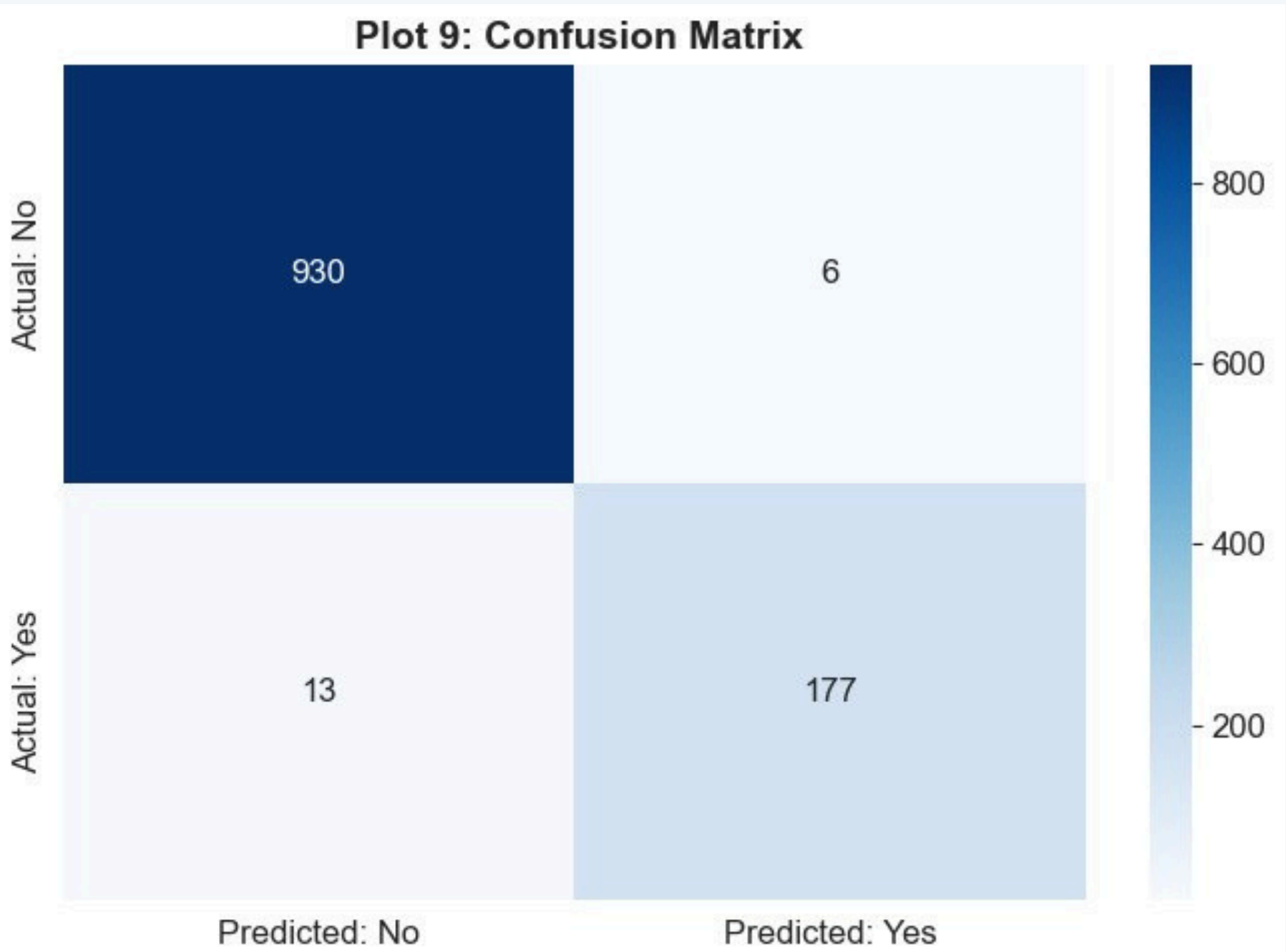
Method: 5-fold cross-validation
Scoring: F1-Score
Total fits: 120

Best Parameters Found:

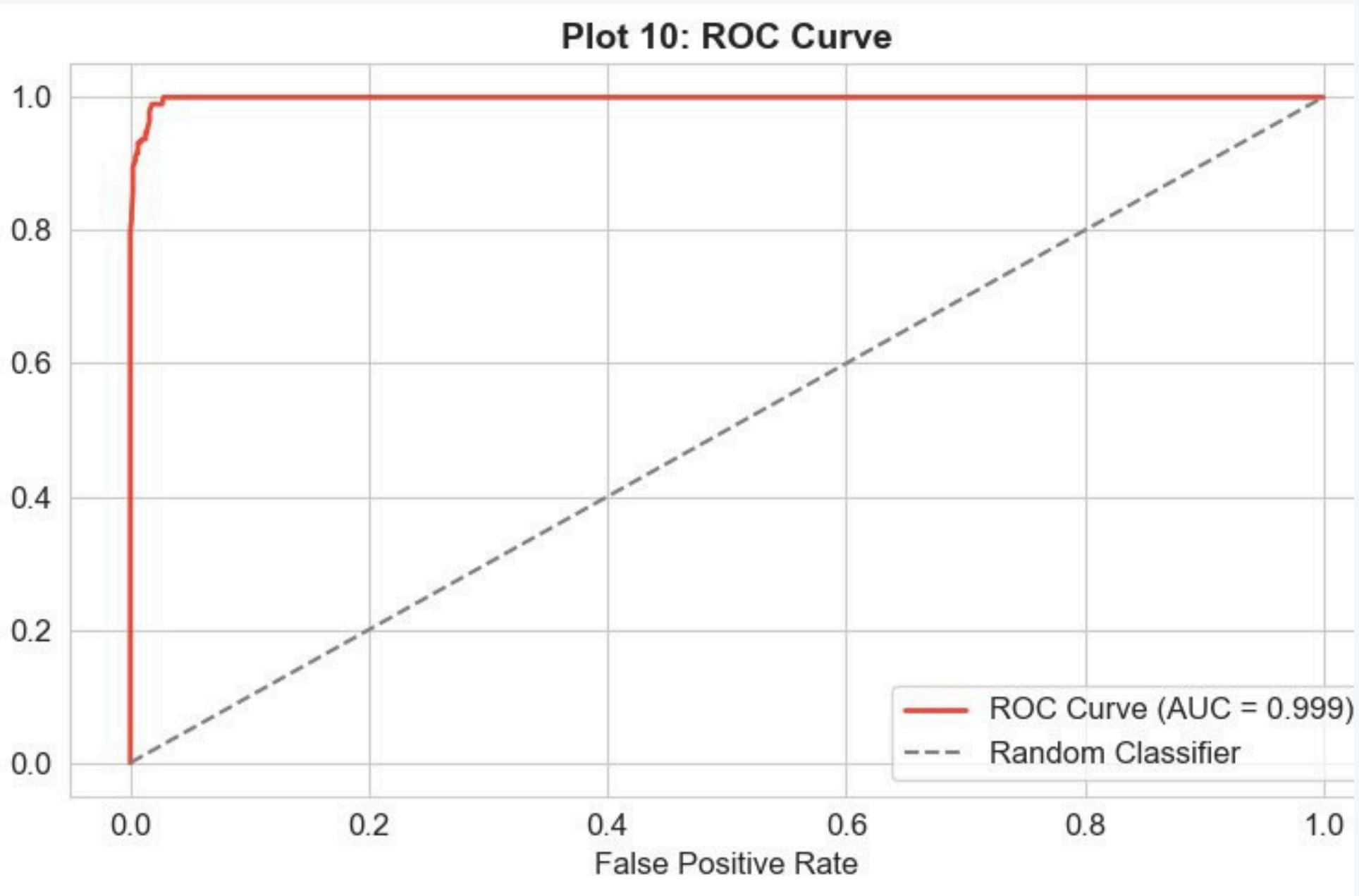
class_weight	balanced
max_depth	None
min_samples_split	2
n_estimators	200

Model Evaluation

Confusion Matrix



ROC Curve (AUC = 0.999)



98.3%

Accuracy

96.7%

Precision

93.2%

Recall

94.9%

F1-Score

0.999

ROC-AUC

Validation — Why It Matters

What is Validation?

Validation is the process of evaluating how well a model performs on data it was never trained on. Without it, a model can memorise the training data perfectly but fail on new real-world data (overfitting).

How We Validated

80/20 train-test split with stratification to keep the same churn ratio in both sets.

5-fold cross-validation during GridSearchCV to ensure parameter choices generalise across different data subsets.

Results

Precision: 96.7% $\geq$ 0.3 requirement

Recall: 93.2% $\geq$ 0.3 requirement

Cross-validation confirmed results are consistent and not due to lucky data splits.

Deployment — Live Web Application

1 Train & Save Model

Random Forest trained in notebook. model.pkl, scaler.pkl, features.pkl saved with joblib.

2 Flask Web App

app.py loads the saved model and serves predictions via a user-friendly HTML form.

3 GitHub Repository

All project files pushed to GitHub including notebook, app, dataset, and pkl files.

4 Railway Deployment

Web app deployed on Railway.app. Live and accessible via a public URL.

App Features

→ Dropdowns for categorical inputs

→ Sliders for numeric inputs

→ 3 clear sections

→ Live churn probability %

→ Hosted on Railway.app

Conclusion

01

Best Model

Tuned Random Forest achieved 98.3% accuracy, 96.7% precision, and 93.2% recall.

02

Key Predictors

Tenure, Complain, EngagementScore, and DaySinceLastOrder were the top features.

03

Business Insight

New customers (low tenure) and customers who complained are at highest churn risk.

THANK YOU